

Module Review: L-Methionine Biosynthesis (from L-homoserine) in *Pseudomonas putida* KT2440

Taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556)

Pathway bucket: kegg:ppu00270 "Cysteine and methionine metabolism" **Module template:** de novo L-methionine from L-homoserine (3 steps, each with route alternatives) **Prepared for:** manual module-satisfiability and gene-annotation curation

1. Executive summary

The three-step L-methionine-from-homoserine module is **SATISFIABLE** in *P. putida* KT2440: every step is covered by at least one credible candidate gene. The organism uses the characteristic *Pseudomonas* route — **O-succinyl activation** of homoserine followed by **single-step direct sulfhydrylation** — rather than the acetyl/trans-sulfuration-only logic assumed by many templates.

Three curation-critical conclusions:

- 1. Activation (step 1) is covered with strong, target-specific evidence.** PP_5097 (metXS, Q88CT3) is a homoserine **O-succinyl**transferase (EC 2.3.1.46), UniProt **PE=1 "evidence at protein level."** Despite the *metX* family name it transfers **succinyl**, not acetyl — a well-documented misannotation hazard (Bastard et al. 2017,

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- 2. Sulfur incorporation (step 2) is covered by direct sulfhydrylation** via PP_2001 (metZ), which is substrate-matched to the O-succinyl intermediate. A parallel forward **trans-sulfuration** route (PP_0659 metB + PP_4348 metC) is also present. Two other PF01053 paralogs (PP_4594, PP_2528/metY) are **paralog-ambiguous** and should not be counted as primary evidence.
- 3. Methylation (step 3): only the cobalamin-DEPENDENT alternative is genuinely present.** MetH (PP_2375) is a complete 1235-aa multidomain enzyme. The lone metE annotation

(PP_2698, 344 aa, PE=4) is a **truncated single-domain protein and a likely over-annotation** — a functional cobalamin-independent MetE (~750 aa, two domains) is absent from the proteome. De novo methionine synthesis in KT2440 is effectively **B12-dependent**.

The module boundary framing "metA=succinyl vs metX=acetyl" and "metE OR metH both realized" is **incorrect for this lineage** and warrants `module_needs_revision` notes (see §6).

2. Target-organism pathway definition

Included process: conversion of L-homoserine → O-succinyl-L-homoserine → L-homocysteine → L-methionine (3 enzymatic steps). This is the terminal, methionine-specific segment of the aspartate/homoserine branch.

Neighboring processes to keep separate (do NOT fold into this module): - Upstream aspartate→homoserine (`thrA`/`hom` PP_1470/PP_0664, `asd` PP_1989, `lysC`/aspartokinase PP_4473, `thrB` PP_0RK8) — threonine/lysine/homoserine supply, KEGG ppu00260/00261/00300. - Cysteine biosynthesis / sulfur assimilation (`cysK` PP_4571, `cysM` PP_1654, `cysE`/serine acetyltransferases) — supplies the cysteine sulfur donor for trans-sulfuration but is a distinct module (ppu00920/00272). - Downstream S-adenosylmethionine cycle: `metK` (PP_4967, SAM synthase), `ahcY` (PP_4976), `metH`-linked methyl-cycle — SAM/SAH metabolism, not de novo Met. - Methionine **catabolism/salvage**: `mdeA` (PP_1308, methionine γ -lyase), the MTA/methylthioribose salvage genes (`mtnA` PP_1766, `PP_3254` nucleosidase), and methionine-sulfoxide reductase (`msrC` PP_1877). These are in the same KEGG map but are not biosynthetic. - The many DNA cytosine-methyltransferases (PP_3029/PP_3912/PP_3989), serine dehydratases (`tdcG`), and aminotransferases in the candidate list are **map co-occurrence artifacts** of ppu00270, not methionine-biosynthesis genes.

Alternate names / database definitions: KEGG module **M00017** (methionine biosynthesis, aspartate → homocysteine → methionine); MetaCyc "L-methionine biosynthesis I/II/III" (I = trans-sulfuration + acetyl; II = direct sulfhydrylation; III = succinyl trans-sulfuration); UniPathway "L-methionine biosynthesis via de novo pathway." GapMind for amino-acid biosynthesis models the same step/route structure used here.

3. Expected step model (as realized in KT2440)

Step	Template alternatives	Realized in KT2440	Status
1. Homoserine activation	O-succinyl (metA) or O-acetyl (metX)	O-succinyl by MetX-family enzyme PP_5097 (EC 2.3.1.46)	covered (direct, PE=1)
2. Sulfur → homocysteine	direct sulfhydrylation (metY/metZ) or trans-sulfuration (metB+metC)	metZ direct sulfhydrylation (PP_2001) on O-succinyl-homoserine; parallel trans-sulfuration (PP_0659 metB + PP_4348 metC)	covered (homology, strong)
3. Homocysteine methylation	cobalamin-independent (metE) or -dependent (metH)	metH only (PP_2375); metE alternative absent/degenerate	covered by metH; metE = candidate_uncertain/gap

Ordering constraints (activation → sulfur → methylation) are satisfied.

4. Candidate genes and evidence

High-confidence, module-defining

PP_5097 **metXS** (Q88CT3) — **Step 1, activation.** Homoserine O-succinyltransferase, EC 2.3.1.46; catalytic reaction L-homoserine + succinyl-CoA → O-succinyl-L-homoserine + CoA; UniPathway step 1/1; **UniProt PE=1 (protein-level evidence)**. Highest-confidence gene in the set. Caveat: **metX/metA** gene names do **not** indicate acyl donor (**P 28581482**);

the succinyl activity here is substrate-consistent with downstream MetZ. → **covered**.

PP_2001 **metZ** (Q88LD4) — **Step 2, direct sulfhydrylation.** O-succinylhomoserine sulfhydrylase, EC 2.5.1.-; O-succinyl-L-homoserine + H₂S → L-homocysteine + succinate; UniPathway L-homocysteine step 1/1; PE=3 (homology). Substrate-matched to PP_5097. Directly

supported by the *P. aeruginosa* precedent (P 7704274).

→ **covered** (transfer from *P. aeruginosa* is strong: same genus, same intermediate).

PP_2375 metH (Q88KB5) — Step 3, methylation. Cobalamin-dependent methionine synthase, EC 2.1.1.13; complete 1235-aa multidomain protein (PF02310 B12-binding, PF02607, PF00809 pterin, PF02965, PF02574); PE=3. → **covered**; makes de novo Met **B12-dependent**.

PP_0659 metB (Q88Q39) — Step 2, trans-sulfuration (cystathionine formation). Cystathionine γ -synthase, PF01053, IPR054542; GO transsulfuration. Forward metB of the parallel route. → **covered (secondary route)**.

PP_4348 metC (Q88EV4) — Step 2, trans-sulfuration (cystathionine cleavage). Cystathionine β -lyase, PF01053 + IPR006233 (MetC-specific); L,L-cystathionine + H₂O → L-homocysteine + pyruvate + NH₄⁺. → **covered (secondary route)**.

Paralog-ambiguous / context-specific (flag, do not use as primary evidence)

PP_4594 (Q88E72) — "cystathionine γ -synthase". PF01053 but GO = **cystathionine γ -LYASE + L-cysteine biosynthesis via cystathionine**; best interpreted as a reverse-transsulfuration enzyme (cysteine formation), not the forward metB. → **candidate_uncertain**.

PP_2528 metY (Q88JW9) — "O-acetylhomoserine (thiol)-lyase". PF01053 + IPR006235; GO O-acetylhomoserine aminocarboxypropyltransferase + cysteine synthase. Its substrate (O-acetyl-homoserine) is **not** the intermediate produced by the O-succinyl activation route, so its de novo methionine role in KT2440 is uncertain (may act in cysteine/sulfide chemistry or on alternative substrates). → **candidate_uncertain**.

PP_2698 metE (Q88JF1) — "5-methyl-THF-triglutamate-homocysteine methyltransferase". Only 344 aa, PE=4 (predicted), carries **only** the C-terminal PF01717 (Meth_synt_2) domain. Functional MetE needs both N-terminal PF08267 and C-terminal PF01717 (~750 aa). → **likely over-annotation / gap** (see §5).

PP_4637 (Q88E31) — MetE-family (274 aa, PF08267 only). Partial N-terminal domain; not a stand-alone methionine synthase. → not a functional metE.

Related but out-of-module (context only)

metK (PP_4967, SAM synthase), ahcY (PP_4976, SAH hydrolase), mdeA (PP_1308, Met γ -lyase = catabolism), mtmA / PP_3254 (Met salvage), Q88Q40/Q88J73 (homocysteine S-methyltransferase/ MmuM-type salvage, not de novo metE).

5. Gaps, ambiguities, and likely over-annotations

- metE step (cobalamin-independent methionine synthase) — likely GAP / over-annotation.** No full-length two-domain MetE in UP000000556. The single `metE` (PP_2698) is a truncated PF01717-only protein (PE=4). This is the most consequential deviation: KT2440's de novo methylation is realized **only** via cobalamin-dependent MetH, making the pathway **B12-dependent**. This is physiologically viable because KT2440 encodes a **complete de novo cobalamin biosynthesis operon set** (`cobI/G/J/M/H/K/L`, `cbiD`, cobaltochelatase `cobN`, adenosyltransferases `cobO/pduO`, `cobQ/D/S/T`, plus a cobalamin ABC transporter — 22 `corrin/B12` genes in UP000000556), so MetH has its cofactor and no functional MetE is required. The "metE" template alternative is therefore **not genuinely satisfied** (not merely unannotated). → mark **candidate_uncertain**, lean **gap**.
- Acyl-donor mis-framing.** The template's "metA=succinyl vs metX=acetyl" dichotomy is wrong here: a MetX-family enzyme performs the **succinyl** activation
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 predicts >60% of MetA/MetX sequences are misannotated). Curators must derive acyl donor from biochemistry/InterPro sub-signature, not gene name.
- PF01053 paralog family (metZ/metB/metC/metY/PP_4594).** Six+ proteins share this PLP fold with broad EC (2.5.1.-, 2.5.1.49) and overlapping GO; specificity must be assigned from InterPro sub-signatures (IPR006233 metC; IPR006235 metY; IPR054542 metB-like) rather than family-level annotation. PP_4594 and PP_2528 are the main over-propagation risks.
- Bucket contamination.** The 46-gene candidate list mixes in DNA methyltransferases, serine dehydratases, sulfurtransferases, and generic aminotransferases that co-occur on KEGG map ppu00270 but are unrelated to methionine biosynthesis.

6. Module and GO-curation recommendations

Module step	Recommended status	Rationale
Homoserine activation (acylation)	covered — O-succinyl branch (PP_5097)	Direct PE=1 evidence; acetyl branch not_expected_in_target_taxon
— O-acetyltransferase (metX-acetyl) sub-branch	not_expected_in_target_taxon	KT2440 activation is succinyl
Sulfur incorporation — direct sulfhydrylation (metZ)	covered (primary)	PP_2001 substrate-matched; genus precedent
Sulfur incorporation — trans-sulfuration (metB+metC)	covered (parallel/secondary)	PP_0659 + PP_4348
— metY (O-acetylhomoserine) direct sulfhydrylation	candidate_uncertain	PP_2528 substrate mismatch
Methylation — cobalamin-dependent (metH)	covered	Complete PP_2375; KT2440 makes B12 de novo (viable)
Methylation — cobalamin-independent (metE)	candidate_uncertain → gap	PP_2698 truncated single-domain, PE=4

Module-boundary revisions (**module_needs_revision**): 1. Reframe step 1 as "homoserine acyl-transfer (acetyl **or** succinyl), donor determined biochemically" rather than binding donor to the metA/metX gene name. 2. Note lineage rule for *Pseudomonas*: **O-succinyl + direct sulfhydrylation (metZ)** is diagnostic and unusual (only genus using O-succinyl-homoserine for direct sulfhydrylation,).

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3. Add a satisfiability caveat that the metE alternative can be a truncated over-annotation; require a length/two-domain (PF08267+PF01717) check before scoring metE "present."

GO-curation: PP_2698 GO 5-methyltetrahydropteroyltriglutamate-homocysteine S-methyltransferase activity should be reviewed/qualified given the truncation (candidate for NOT/with-caveat). PP_4594 GO conflicts (γ -synthase name vs γ -lyase GO) should be reconciled. No new GO terms appear necessary; existing terms suffice if applied at correct specificity.

7. Genes to promote to full `fetch-gene` review

1. **PP_2698 `metE` (Q88JF1)** — resolve whether truncated protein is a functional methionine synthase, a salvage-type homocysteine methyltransferase, or a pseudo/fragment. Highest priority (decides `metE` gap call).
2. **PP_5097 `metXS` (Q88CT3)** — confirm the experimental basis of PE=1 and the succinyl-CoA specificity (anchor gene; verify it is the KT2440 activation enzyme).
3. **PP_2001 `metZ` (Q88LD4)** — confirm O-succinylhomoserine (vs O-acetyl) specificity in KT2440; central to the direct-sulphydrylation call.
4. **PP_4594 (Q88E72) and PP_2528 `metY` (Q88JW9)** — assign forward-`metB` vs reverse- γ -lyase (PP_4594) and de-novo relevance vs cysteine/O-acetyl chemistry (`metY`).

8. Key references

- Foglino M, et al. (1995) *A direct sulphydrylation pathway is used for methionine biosynthesis in Pseudomonas aeruginosa.*

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Establishes O-succinylhomoserine synthase + `metZ` direct sulphydrylation; *Pseudomonas* uniquely uses O-succinyl-homoserine for direct sulphydrylation; trans-sulfuration present in parallel.

- Bastard K, et al. (2017) *Parallel evolution of non-homologous isofunctional enzymes in methionine biosynthesis.* Nat Chem Biol.

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MetA/MetX acyl-donor cannot be inferred from family; >60% of ~10,000 sequences likely misannotated; recent bacteria evolved succinyl-using MetX.

- Bourhy P, et al. (1997) *Homoserine O-acetyltransferase ... Leptospira meyeri ... MetX*.

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Illustrates a MetX with acetyl activity (contrasting lineage), reinforcing donor ambiguity.

- UniProt (2024) entries Q88CT3, Q88LD4, Q88Q39, Q88EV4, Q88E72, Q88JW9, Q88JF1, Q88E31, Q88KB5 (proteome UP000000556) — annotations, EC, Pfam/InterPro, protein-existence evidence used throughout.

Evidence transfer notes: Step-1 activation is **direct** (PE=1, KT2440). Steps 2–3 gene assignments are **homology-based** within KT2440 but strongly corroborated by **same-genus** experimental work (P. aeruginosa,

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genus transfer is strong for the succinyl/direct-sulfhydrylation logic. The metE gap call rests on **sequence/domain-architecture** analysis of the KT2440 proteome (strong), general MetE biochemistry (strong), and the presence of a **complete de novo cobalamin biosynthetic gene set** (strong) that makes a MetH-only route viable. Remaining open experimental questions: a direct growth demonstration of B12-conditional methionine auxotrophy in a KT2440 Δ metH background, and substrate-specificity assays for PP_2528 (metY) and PP_4594 to settle their de novo relevance.